Defect Predictor Wrap Up

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Optimization Results

Absence Testing - remove a factor for each set

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--- Phase 2: Testing Weight Sets ---

Testing Weights (Set 1): {'w1': 0.0, 'w2': 0.2, 'w3': 0.2, 'w4': 0.2, 'w5': 0.2, 'w6': 0.2}
    Accuracy for config1.json: 0.50
    Accuracy for config2.json: 0.50
    Accuracy for config3.json: 0.20
    Accuracy for config4.json: 0.50

Average Accuracy for Weight Set 1: 0.42

Testing Weights (Set 2): {'w1': 0.2, 'w2': 0.0, 'w3': 0.2, 'w4': 0.2, 'w5': 0.2, 'w6': 0.2}
    Accuracy for config1.json: 0.50
    Accuracy for config2.json: 0.50
    Accuracy for config3.json: 0.50
    Accuracy for config4.json: 0.50

Average Accuracy for Weight Set 2: 0.50

Testing Weights (Set 3): {'w1': 0.2, 'w2': 0.2, 'w3': 0.0, 'w4': 0.2, 'w5': 0.2, 'w6': 0.2}
    Accuracy for config1.json: 0.50
    Accuracy for config2.json: 0.50
    Accuracy for config3.json: 0.50
    Accuracy for config4.json: 0.67

Average Accuracy for Weight Set 3: 0.54

Testing Weights (Set 4): {'w1': 0.2, 'w2': 0.2, 'w3': 0.2, 'w4': 0.0, 'w5': 0.2, 'w6': 0.2}
    Accuracy for config1.json: 1.00
    Accuracy for config2.json: 0.20
    Accuracy for config3.json: 0.50
    Accuracy for config4.json: 0.67

Average Accuracy for Weight Set 4: 0.59

Testing Weights (Set 5): {'w1': 0.2, 'w2': 0.2, 'w3': 0.2, 'w4': 0.2, 'w5': 0.0, 'w6': 0.2}
    Accuracy for config1.json: 1.00
    Accuracy for config2.json: 0.20
    Accuracy for config3.json: 0.50
    Accuracy for config4.json: 0.50

Average Accuracy for Weight Set 5: 0.55

Testing Weights (Set 6): {'w1': 0.2, 'w2': 0.2, 'w3': 0.2, 'w4': 0.2, 'w5': 0.2, 'w6': 0.0}
    Accuracy for config1.json: 0.50
    Accuracy for config2.json: 0.50
    Accuracy for config3.json: 0.50
    Accuracy for config4.json: 0.50

Average Accuracy for Weight Set 6: 0.50


--- Optimization Complete ---

Summary of Average Accuracies per Weight Set:
  Weights: {'w1': 0.2, 'w2': 0.2, 'w3': 0.2, 'w4': 0.0, 'w5': 0.2, 'w6': 0.2} -> Avg Accuracy: 0.59
  Weights: {'w1': 0.2, 'w2': 0.2, 'w3': 0.2, 'w4': 0.2, 'w5': 0.0, 'w6': 0.2} -> Avg Accuracy: 0.55
  Weights: {'w1': 0.2, 'w2': 0.2, 'w3': 0.0, 'w4': 0.2, 'w5': 0.2, 'w6': 0.2} -> Avg Accuracy: 0.54
  Weights: {'w1': 0.2, 'w2': 0.0, 'w3': 0.2, 'w4': 0.2, 'w5': 0.2, 'w6': 0.2} -> Avg Accuracy: 0.50
  Weights: {'w1': 0.2, 'w2': 0.2, 'w3': 0.2, 'w4': 0.2, 'w5': 0.2, 'w6': 0.0} -> Avg Accuracy: 0.50
  Weights: {'w1': 0.0, 'w2': 0.2, 'w3': 0.2, 'w4': 0.2, 'w5': 0.2, 'w6': 0.2} -> Avg Accuracy: 0.42

Best Performing Weight Set:
  Weights: {'w1': 0.2, 'w2': 0.2, 'w3': 0.2, 'w4': 0.0, 'w5': 0.2, 'w6': 0.2}
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Extreme Value Testing - Weight one factor higher than the rest for each set

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--- Phase 2: Testing Weight Sets ---

Testing Weights (Set 1): {'w1': 0.5, 'w2': 0.1, 'w3': 0.1, 'w4': 0.1, 'w5': 0.1, 'w6': 0.1}
    Accuracy for config1.json: 1.00
    Accuracy for config2.json: 0.50
    Accuracy for config3.json: 1.00
    Accuracy for config4.json: 0.50

Average Accuracy for Weight Set 1: 0.75

Testing Weights (Set 2): {'w1': 0.1, 'w2': 0.5, 'w3': 0.1, 'w4': 0.1, 'w5': 0.1, 'w6': 0.1}
    Accuracy for config1.json: 1.00
    Accuracy for config2.json: 0.20
    Accuracy for config3.json: 0.50
    Accuracy for config4.json: 0.50

Average Accuracy for Weight Set 2: 0.55

Testing Weights (Set 3): {'w1': 0.1, 'w2': 0.1, 'w3': 0.5, 'w4': 0.1, 'w5': 0.1, 'w6': 0.1}
    Accuracy for config1.json: 0.50
    Accuracy for config2.json: 0.20
    Accuracy for config3.json: 0.20
    Accuracy for config4.json: 0.50

Average Accuracy for Weight Set 3: 0.35

Testing Weights (Set 4): {'w1': 0.1, 'w2': 0.1, 'w3': 0.1, 'w4': 0.5, 'w5': 0.1, 'w6': 0.1}
    Accuracy for config1.json: 0.50
    Accuracy for config2.json: 0.20
    Accuracy for config3.json: 0.20
    Accuracy for config4.json: 0.50

Average Accuracy for Weight Set 4: 0.35

Testing Weights (Set 5): {'w1': 0.1, 'w2': 0.1, 'w3': 0.1, 'w4': 0.1, 'w5': 0.5, 'w6': 0.1}
    Accuracy for config1.json: 0.50
    Accuracy for config2.json: 0.50
    Accuracy for config3.json: 0.20
    Accuracy for config4.json: 0.50

Average Accuracy for Weight Set 5: 0.42

Testing Weights (Set 6): {'w1': 0.1, 'w2': 0.1, 'w3': 0.1, 'w4': 0.1, 'w5': 0.1, 'w6': 0.5}
    Accuracy for config1.json: 0.50
    Accuracy for config2.json: 0.20
    Accuracy for config3.json: 0.20
    Accuracy for config4.json: 0.50

Average Accuracy for Weight Set 6: 0.35


--- Optimization Complete ---

Summary of Average Accuracies per Weight Set:
  Weights: {'w1': 0.5, 'w2': 0.1, 'w3': 0.1, 'w4': 0.1, 'w5': 0.1, 'w6': 0.1} -> Avg Accuracy: 0.75
  Weights: {'w1': 0.1, 'w2': 0.5, 'w3': 0.1, 'w4': 0.1, 'w5': 0.1, 'w6': 0.1} -> Avg Accuracy: 0.55
  Weights: {'w1': 0.1, 'w2': 0.1, 'w3': 0.1, 'w4': 0.1, 'w5': 0.5, 'w6': 0.1} -> Avg Accuracy: 0.42
  Weights: {'w1': 0.1, 'w2': 0.1, 'w3': 0.1, 'w4': 0.1, 'w5': 0.5, 'w6': 0.1} -> Avg Accuracy: 0.42
  Weights: {'w1': 0.1, 'w2': 0.1, 'w3': 0.5, 'w4': 0.1, 'w5': 0.1, 'w6': 0.1} -> Avg Accuracy: 0.35
  Weights: {'w1': 0.1, 'w2': 0.1, 'w3': 0.1, 'w4': 0.5, 'w5': 0.1, 'w6': 0.1} -> Avg Accuracy: 0.35
  Weights: {'w1': 0.1, 'w2': 0.1, 'w3': 0.1, 'w4': 0.1, 'w5': 0.1, 'w6': 0.5} -> Avg Accuracy: 0.35

Best Performing Weight Set:
  Weights: {'w1': 0.5, 'w2': 0.1, 'w3': 0.1, 'w4': 0.1, 'w5': 0.1, 'w6': 0.1}
  Highest Average Accuracy: 0.75
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Revisit Factors in the Formula

- Recency (Date committed): 22.5% -> 45%
- Frequency (How often the files are changed): 30% -> 17.5%
- Number of lines modified: 15% -> 7.5%
    - Lines added + lines deleted
- How many lines of code (loc) in the file: 10% -> 7.5%
- Cyclomatic Complexity: 12.5% -> 10%
- Number of minor contributors: 10% -> 12.5%

Results

- Recency of change is the most significant factor
  - Accuracy increased from .66 to .74 after increasing weight